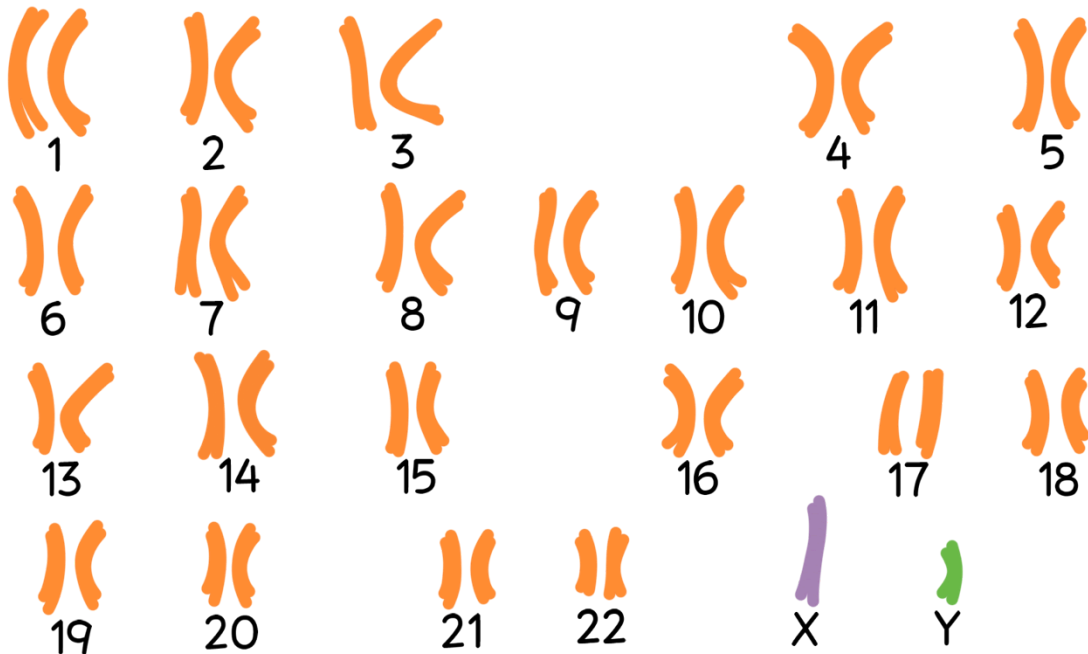
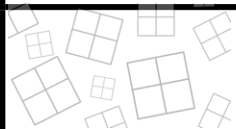


## Amoeba Sisters Video Recap: Punnett Squares and Sex-Linked Traits



1. How many chromosomes in total are there in this individual's karyotype? \_\_\_\_\_
2. How many chromosome pairs are there? (Note: count the sex chromosomes as a pair!) \_\_\_\_\_
3. Circle the sex chromosomes. How many sex chromosomes are there? \_\_\_\_\_
4. How many autosomes (chromosomes that aren't sex chromosomes) are there? \_\_\_\_\_
5. Is this a general karyotype of a male or female? \_\_\_\_\_



For the following questions, consider a trait that is sex-linked recessive and on the X chromosome. The letter "b" will be used for this allele.

6. Circle the genotype(s) that would generally denote a **female who has this sex-linked recessive trait**.

BB      Bb      bb       $X^B X^B$        $X^B Y^B$        $X^B X^b$        $X^b Y$        $X^b X^b$        $X^b Y^b$        $X^B Y$

7. Circle the genotype(s) that would generally denote a **female who carries the sex-linked allele but does not have the sex-linked recessive trait**.

BB      Bb      bb       $X^B X^B$        $X^B Y^B$        $X^B X^b$        $X^b Y$        $X^b X^b$        $X^b Y^b$        $X^B Y$

8. Circle the genotype(s) that would generally denote a **female who neither has the sex-linked recessive trait nor carries the allele**.

BB      Bb      bb       $X^B X^B$        $X^B Y^B$        $X^B X^b$        $X^b Y$        $X^b X^b$        $X^b Y^b$        $X^B Y$

9. Circle the genotype(s) that would generally denote a **male who has the sex-linked recessive trait**.

BB      Bb      bb       $X^B X^B$        $X^B Y^B$        $X^B X^b$        $X^b Y$        $X^b X^b$        $X^b Y^b$        $X^B Y$

10. Circle the genotype(s) that would generally denote a **male who does not have the sex-linked trait**.

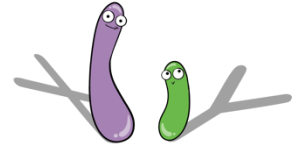
BB      Bb      bb       $X^B X^B$        $X^B Y^B$        $X^B X^b$        $X^b Y$        $X^b X^b$        $X^b Y^b$        $X^B Y$



## Amoeba Sisters Video Recap: Punnett Squares and Sex-Linked Traits

For each statement below, write "true" or "false." If false, please *correct* the statement underneath.

11. \_\_\_\_\_ Many traits are not sex-linked and are instead on **autosomes**. In Punnett squares that are using alleles on autosomes, you will *not* put sex chromosomes around the Punnett square.



12. \_\_\_\_\_ Sex-linked *recessive* traits are generally more common in males than females.

13. \_\_\_\_\_ Most Punnett square problems using sex-linked traits will involve genes on the Y chromosomes.

14. Red-green colorblindness is a sex-linked recessive trait. Show in the Punnett square how a male who has this condition could have biological parents who both do not.

|  |  |
|--|--|
|  |  |
|  |  |

Using the filled in Punnett square, *explain* how a male who has this condition could have biological parents who both do not?

15. Complete the boxes of the Punnett square below.

|          |          |          |
|----------|----------|----------|
|          | <b>X</b> | <b>Y</b> |
| <b>X</b> |          |          |
| <b>X</b> |          |          |

Using the filled in Punnett square, explain whether it is the male or female parent who generally determines the sex of the baby?

16. Some disorders are sex-linked dominant on the X chromosome. Dominant disorders generally only need one dominant allele present. Using the allele "D" on the X chromosome, show a cross with a female who does not have a dominant sex-linked disorder ( $X^dX^d$ ) with a male who does.

|  |  |
|--|--|
|  |  |
|  |  |

In this particular cross, was this condition more common in female or male offspring? Why?

